CLASS 11 MATHEMATICS

Chapter 1: COMPLEX NUMBERS — Complete Study Notes

Definitions • Properties • Formulas • Worked Examples • Revision Sheet

Note: These notes cover the main topics visible in your video and the standard Class 11 Complex Numbers chapter. Match notation and exercise numbers with your own textbook.

1. Number System and Complex Numbers

A **number system** is a collection of numbers. Common sets are natural numbers (N), whole numbers (W), integers (Z), rational numbers (Q), irrational numbers, and real numbers (R).

A **complex number** is a number of the form **z = a + ib**, where a and b are real numbers and **i =** $\sqrt{-1}$. Here, a is the real part and b is the imaginary part.

$i^2 = -1$, $i^3 = -i$, $i^4 = 1$

The powers of i repeat after every four powers: i, –1, –i, 1, i, –1, –i, 1, …

Examples

$z = 5 + 3i \Rightarrow Re(z) = 5, Im(z) = 3$

$z = -4 - 7i \Rightarrow Re(z) = -4, Im(z) = -7$

$z = 6 \Rightarrow Re(z) = 6, Im(z) = 0$ (a real number)

$z = -2i \Rightarrow Re(z) = 0, Im(z) = -2$ (a pure imaginary number)

Important: In z = a + ib, the imaginary part is the real coefficient b, not ib.

2. Types / Classification of Complex Numbers

- **Real number:** b = 0, for example 7 = 7 + 0i.
- **Pure imaginary number:** a = 0 and $b \neq 0$, for example 4i or –3i.
- **Non-real complex number:** $b \neq 0$, for example 2 + 5i.
- Every real number is also a complex number, but not every complex number is real.

The set of complex numbers is commonly written as **C**.

3. Equality of Complex Numbers

Two complex numbers are equal if and only if their real parts are equal and their imaginary coefficients are equal.

If a + ib = c + id, then a = c and b = d.

Example

If 3x + i(2y) = 12 + 8i, then 3x = 12 and 2y = 8.

Therefore x = 4 and y = 4.

4. Algebra of Complex Numbers

A. Addition

Add real parts together and imaginary coefficients together.

$(a + ib) + (c + id) = (a + c) + i(b + d)$

Example: $(3 + 2i) + (5 – 7i) = 8 – 5i$

B. Subtraction

$(a + ib) – (c + id) = (a – c) + i(b – d)$

Example: $(6 + 4i) – (2 + 9i) = 4 – 5i$

C. Multiplication

Multiply as ordinary binomials, then replace i^2 by $–1$.

$(a + ib)(c + id) = (ac – bd) + i(ad + bc)$

Example: $(2 + 3i)(4 – i) = 8 – 2i + 12i – 3i^2 = 11 + 10i$

D. Division

To divide complex numbers, multiply numerator and denominator by the conjugate of the denominator.

$(a + ib)/(c + id) = [(a + ib)(c – id)]/(c^2 + d^2)$, where $c^2 + d^2 \neq 0$

Example: $(1 + 2i)/(3 – i) = [(1 + 2i)(3 + i)]/[(3 – i)(3 + i)]$

$= (1 + 7i)/10 = 1/10 + (7/10)i$

5. Conjugate of a Complex Number

If $z = a + ib$, its **complex conjugate** is written $z\blacksquare$ (or z^*) and is $z\blacksquare = a – ib$. The sign of the imaginary term changes.

Example: $z = 4 – 9i \Rightarrow z\blacksquare = 4 + 9i$

Important conjugate properties

$z + z\blacksquare = 2\,\mathrm{Re}(z) = 2a$

$z – z\blacksquare = 2ib$

$z \cdot z\blacksquare = a^2 + b^2$ (a non-negative real number)

overline($z\blacksquare + z\blacksquare$) = $z\blacksquare\blacksquare + z\blacksquare\blacksquare$

overline($z\blacksquare z\blacksquare$) = $z\blacksquare\blacksquare\ z\blacksquare\blacksquare$

A complex number is real exactly when $z = z\blacksquare$.

6. Modulus / Absolute Value

For $z = a + ib$, the **modulus** (or absolute value) is the distance of the point (a, b) from the origin in the complex plane.

$|z| = \sqrt{(a^2 + b^2)}$

Example: $z = 3 + 4i \Rightarrow |z| = \sqrt{(3^2 + 4^2)} = \sqrt{25} = 5$

Properties of modulus

$|z| \geq 0$; and $|z| = 0$ if and only if $z = 0$

$|z\blacksquare| = |z|$

$|z\blacksquare z\blacksquare| = |z\blacksquare||z\blacksquare|$

$|z\blacksquare/z\blacksquare| = |z\blacksquare|/|z\blacksquare|$, when $z\blacksquare \neq 0$

$|z\blacksquare + z\blacksquare| \leq |z\blacksquare| + |z\blacksquare|$ (triangle inequality)

$z\ z\blacksquare = |z|^2$

7. Geometrical Representation: Argand / Complex Plane

A complex number $z = a + ib$ is represented by the point $P(a, b)$. The horizontal axis is the **real axis** and the vertical axis is the **imaginary axis**. This coordinate diagram is called the Argand plane or complex plane.

- x-coordinate = real part a.
- y-coordinate = imaginary coefficient b.
- Distance $OP = |z| = \sqrt{(a^2 + b^2)}$.

The argument (angle) of z is the angle θ made by OP with the positive real axis, measured according to the convention used in the textbook. For $z \neq 0$, $\tan \theta = b/a$ when $a \neq 0$; the quadrant must be checked before choosing θ.

A common polar form is **$z = r(\cos \theta + i \sin \theta)$**, where $r = |z|$ and θ is an argument of z. The principal argument is often written $\text{Arg}(z)$ and is chosen in a specified interval, commonly $(-\pi, \pi]$ or $[0, 2\pi)$, depending on the textbook.

8. Additive and Multiplicative Identities / Inverses

- **Additive identity:** 0, because $z + 0 = z$.
- **Multiplicative identity:** 1, because $z \times 1 = z$.
- **Additive inverse:** $-z$. For $z = a + ib$, $-z = -a - ib$, and $z + (-z) = 0$.
- **Multiplicative inverse:** $1/z$ for $z \neq 0$.

$$1/(a + ib) = (a - ib)/(a^2 + b^2), \text{ provided } z \neq 0$$

Example: $1/(2 + i) = (2 - i)/(2^2 + 1^2) = (2 - i)/5$

9. Basic Properties of Complex Numbers

For complex numbers z_1, z_2, z_3:

Commutative: $z_1 + z_2 = z_2 + z_1$; $z_1 z_2 = z_2 z_1$

Associative: $(z_1 + z_2) + z_3 = z_1 + (z_2 + z_3)$

$(z_1 z_2) z_3 = z_1 (z_2 z_3)$

Distributive: $z_1(z_2 + z_3) = z_1 z_2 + z_1 z_3$

Complex numbers form a field under addition and multiplication: addition, subtraction, multiplication, and division by a non-zero complex number stay within the complex numbers.

10. Solving Quadratic Equations with Complex Roots

A quadratic equation has the form $ax^2 + bx + c = 0$, with $a \neq 0$. Its roots are given by the quadratic formula:

$$x = [-b \pm \sqrt{(b^2 - 4ac)}]/(2a)$$

If the discriminant $D = b^2 - 4ac$ is negative, the square root involves i because $\sqrt{(-k)} = i\sqrt{k}$ for $k > 0$.

Example

$$x^2 + 4 = 0 \Rightarrow x^2 = -4 \Rightarrow x = \pm\sqrt{(-4)} = \pm 2i$$

Example: $x^2 - 2x + 5 = 0$

$$x = [2 \pm \sqrt{(4 - 20)}]/2 = [2 \pm \sqrt{(-16)}]/2 = 1 \pm 2i$$

For a quadratic equation with real coefficients, non-real complex roots occur in conjugate pairs.

11. Worked Examples for Practice

1. Simplify $i^{2\blacksquare}$.
Answer: $27 \div 4$ leaves remainder 3, so $i^{2\blacksquare} = i^3 = -i$.

2. Find the conjugate and modulus of z = –5 + 12i.
Answer: Conjugate = –5 – 12i; modulus = √(25 + 144) = 13.

3. Calculate (4 + i) + (–7 + 6i).
Answer: –3 + 7i.

4. Calculate (3 – 2i)(1 + 4i).
Answer: 3 + 12i – 2i – 8i² = 11 + 10i.

5. Find 1/(1 – 3i).
Answer: (1 + 3i)/(1 + 9) = (1 + 3i)/10.

6. If z = a + ib and z■ = 6 – 2i, find z.
Answer: z = 6 + 2i.

7. Find x and y if (x + 2) + i(3y – 1) = 7 + 8i.
Answer: x + 2 = 7 ⇒ x = 5; 3y – 1 = 8 ⇒ y = 3.

8. Solve x² + 9 = 0.
Answer: x = ±3i.

12. Common Mistakes to Avoid

1. Do not forget that i² = –1, not +1.
2. In a + ib, Im(z) = b, not ib.
3. For division, use the conjugate of the denominator.
4. |a + ib| = √(a² + b²); modulus is never negative.
5. When finding an argument, check the quadrant; tan■¹(b/a) alone may give the wrong angle.
6. Division by zero is not defined; the multiplicative inverse of 0 does not exist.

13. Quick Formula Sheet

Topic	Formula / fact
Definition	z = a + ib; i² = –1
Real / imaginary parts	Re(z) = a; Im(z) = b
Powers of i	i■ = 1, i¹ = i, i² = –1, i³ = –i; repeat every 4
Addition	(a + ib) + (c + id) = (a+c) + i(b+d)
Multiplication	(a+ib)(c+id) = (ac–bd) + i(ad+bc)
Conjugate	z■ = a – ib
Modulus	|z| = √(a²+b²)
Conjugate product	z z■ = |z|² = a²+b²
Reciprocal	1/(a+ib) = (a–ib)/(a²+b²), z ≠ 0
Polar form	z = r(cos θ + i sin θ), r = |z|
Quadratic formula	x = [–b ± √(b²–4ac)]/(2a)

14. Self-Test MCQs

1. The value of i² is:
A) 1 B) –1 C) i D) –i

2. If z = 3 – 4i, then Re(z) is:
A) 3 B) –4 C) 4 D) –3

3. The conjugate of 2 + 7i is:
A) –2 + 7i B) 2 – 7i C) –2 – 7i D) 7 + 2i

4. The modulus of 5 + 12i is:
A) 7 B) 12 C) 13 D) 17

5. i■ equals:
A) 1 B) i C) –1 D) –i

6. The multiplicative identity is:
A) 0 B) i C) 1 D) –1

7. (1+i)(1–i) equals:
A) 0 B) 1 C) 2 D) 2i

8. The reciprocal of a complex number z is defined when:
A) z=0 B) z≠0 C) z is imaginary only D) always

Answer key: 1-B, 2-A, 3-B, 4-C, 5-D, 6-C, 7-C, 8-B